

Adelowo Adebajji Oluwatimileyin

Data Scientist · Machine Learning Engineer

adelowooluwatimileyin@gmail.com · +39 388 751 5527 · L'Aquila, Italy
github.com/AdebajjiAdelowo · adebanjioluwatimileyin.github.io

PROFILE

Data Scientist and Machine Learning Engineer with 3+ years of industry experience in predictive modelling, operational analytics, signal processing, and deployed ML services. Develops Python and SQL workflows and integrates models into APIs and dashboards. Combines applied mathematical training with controlled model evaluation, statistical analysis, and reproducible experimentation.

PROFESSIONAL EXPERIENCE

Data Scientist | *Intuos Srl*, L'Aquila, Italy Dec 2024 – Present

- Develop flight-phase classifiers and analyse 278,000+ telemetry records in IBM DB2 to support internal flight-performance monitoring.
- Design FFT-based signal-processing pipelines for aircraft engine RPM estimation, frequency analysis, and anomaly detection.
- Build telemetry-processing and operational analytics workflows that turn recorded flight data into model inputs and engineering metrics for internal monitoring.
- Build a Dockerised dashboard using FastAPI, IBM DB2, and React, exposing flight-phase predictions, engineering metrics, and deterministic safety alarms to internal users.

Machine Learning Engineer | *N-and Italia Srl*, Italy Apr 2023 – Nov 2024

- Developed predictive models for vending-machine delivery-failure forecasting using large-scale transaction and telemetry data.
- Performed exploratory and statistical analysis to identify patterns affecting performance and operational reliability.
- Built data-processing and KPI-monitoring workflows to support operational decision-making.

Operational Data Analyst | *eResult Srl*, Cesena, Italy Jan 2023 – Mar 2023

- Optimised SQL queries and automated recurring business-intelligence reporting workflows.
- Transformed operational data into reproducible performance metrics and actionable reporting outputs.

Digital Engineering Intern | *Bridgestone EMIA*, Rome, Italy Apr 2022 – Oct 2022

- Developed computational-geometry algorithms and improved data-structure operations for tire cross-section simulation and finite-element meshing workflows.

Machine Learning Intern | *Fellowship.AI*, Remote May 2022 – Aug 2022

- Implemented an InsetGAN architecture for whole-body image generation and deployed model inference through FastAPI for integration with an iOS application.

TECHNICAL SKILLS

Programming & data: Python, SQL, C++, MATLAB, NumPy, SciPy, Pandas, PostgreSQL, IBM DB2

Machine learning: PyTorch, scikit-learn, TensorFlow, LightGBM; time series, signal processing, computer vision

ML applications: FastAPI, Docker, Git, MLflow, React, REST APIs, RAG, LangChain, NL-to-SQL

Evaluation: held-out and multi-seed experiments, classical baselines, distribution-shift studies

INDEPENDENT RESEARCH PROJECTS

Applied Mathematics & Scientific Machine Learning Jan 2026 – Present

Develop and verify numerical solvers and ML surrogates using convergence studies, analytical solutions, gradient checks, and held-out experiments. Study model accuracy, runtime, and generalisation in PDE simulation and computational imaging.

EDUCATION

M.Sc. Mathematical Engineering | University of L'Aquila, Italy 2018 – 2021

Thesis: *Bounds on Mixing of Passive Scalars Advected by Incompressible Energy- and Enstrophy-Constrained Flows and Anomalous Dissipation.*

Relevant study: machine learning, scientific computing, optimisation, numerical analysis, finite-element methods, and PDEs.

B.Sc. Mathematics | Obafemi Awolowo University, Nigeria 2013 – 2018

Thesis: *Application of Galerkin Finite Element Method to the One-Dimensional Stefan Problem.*

SELECTED PROJECTS

Flight Data Monitoring Dashboard

- Built a FastAPI / IBM DB2 backend and React/Vite frontend, packaged with Docker, for internal analysis of recorded aircraft telemetry.
- Integrated flight-phase predictions, engineering metrics, and deterministic safety alarms for internal users; recorded telemetry is ingested through an external ETL workflow.

Photoacoustic Reconstruction

- Built a synthetic sparse-view imaging pipeline comparing calibrated time reversal, U-Net refinement, and Tikhonov inversion.
- Across 400 held-out images and five networks, U-Net refinement improved PSNR by 4.9 dB (16 sensors) and 3.2 dB (64 sensors) over time reversal; evaluated reconstruction under controlled sound-speed variations.

Navier–Stokes Simulation and Reduced-Order Models

- Built a Fourier pseudo-spectral solver and tensor POD–Galerkin surrogate; verified fourth-order temporal and spectral spatial convergence against analytical and manufactured solutions.
- At rank 16, achieved $38.5\times$ online speed-up at $< 0.1\%$ mean state error on the training trajectory.

Classical vs. Learned PDE Surrogates

- Benchmarked POD–Galerkin, DEIM, MLP, and Fourier Neural Operator surrogates for viscous Burgers' equation.
- On 40 in-distribution cases, POD reached 0.152% relative L^2 error versus 0.90% for FNO (five seeds); evaluated transfer to an unseen initial-condition family to characterise generalisation.

Ensemble Kalman Filtering for Chaotic State Estimation

- Implemented stochastic EnKF with covariance localisation and inflation for Lorenz-96; verified against an exact linear-Gaussian Kalman-filter solution.
- In a 10-seed study with 20 ensemble members and a selected localisation radius, reduced analysis RMSE from 4.63 ± 0.13 to 0.41 ± 0.03 .

LLM Applications and Model Compression

- Implemented medical retrieval-augmented generation with LangChain/ChromaDB and a two-stage enterprise natural-language-to-SQL pipeline.
- Implemented FinBERT layer removal and knowledge distillation, reducing the model from 110M to 88M parameters.

S4D vs. Transformer: Controlled Sequence Modelling

- Implemented S4D and causal attention from scratch in a shared PyTorch backbone; numerically checked convolutional and recurrent S4D equivalence.
- In a fixed-budget, single-seed TinyShakespeare benchmark at context 1024 on a Tesla T4, measured approximately $4.1\times$ training throughput and 1.35 versus 6.46 GiB peak memory relative to a Transformer using plain, unfused attention.

CERTIFICATIONS

IBM Data Science Professional Certificate (IBM / Coursera, 2020); Machine Learning (Stanford University / Coursera, 2020).

SELECTED AWARDS

Professor S.S. Okoya Prize for Excellence in Applied Mathematics (2018); University of L'Aquila International Scholarship (2018–2021).